

Document 2: Extended IEEE Technical Methodology

Project Name: SkyGuard AI

Focus: Architectural Implementation, Edge-to-Cloud Distributed Topologies, and Algorithm Specifications

Abstract

The deployment of machine learning algorithms in national meteorological networks is traditionally bottlenecked by severe computational and bandwidth constraints at the edge. Automatic Weather Stations (AWS) operate in isolated environments, transmitting data via narrow-band cellular or satellite uplinks. To bridge the gap between complex AI anomaly detection and edge hardware realities, SkyGuard AI introduces a novel, distributed 6-Tier Arbitration Architecture.

This extended methodology document details the exact software engineering, computational physics, and topological design patterns required to execute the SkyGuard engine in real-time (<200ms latency). By strategically distributing deterministic physical laws (Tiers 0-3) to the edge microcontrollers, and reserving computationally heavy Isolation Forests and game-theoretic SHAP explainability (Tiers 4-5) for the cloud, the system achieves an unprecedented 83.9% precision rate without overwhelming cellular telemetry networks. Furthermore, this document dissects the core StateManager engine, explaining how strict T-1 data freezing prevents historical data pollution-a critical requirement for sequential probability tests.

I. The Distributed Edge-to-Cloud Network Topology

A. The Bandwidth Bottleneck in Modern Meteorology

Modern Automatic Weather Stations (AWS) are equipped with high-resolution digital transducers capable of sampling the atmosphere at 1Hz (one reading per second). However, transmitting 1Hz 49-feature telemetry over 3G/4G cellular networks or Iridium satellite constellations is financially ruinous for a national weather agency operating 500+ stations. Consequently, data is typically aggregated on the edge logger into 10-minute or 1-hour averages before transmission. This aggregation destroys the high-frequency variance signatures required to detect transient hardware spikes or rapid calibration loss. If an AI is going to detect anomalies, it must have access to the raw, unaggregated 1Hz signal.

B. The Edge Computing Paradigm

To solve this, SkyGuard AI employs a strict Edge Computing topology. The AWS data loggers (typically ARM Cortex-M4 or M7 microcontrollers) are flashed with a lightweight, compiled version of Tiers 0 through 3. * **Tier 0 (Hardware Limits) & Tier 1 (Spike Logic):** Execute in $O(1)$ time complexity directly on the C-based firmware. * **Tier 2 (CUSUM Accumulators):** Requires a tiny memory footprint (tracking only S_t^+ and S_t^- float variables per sensor). * **Tier 3 (Mahalanobis Covariance):** The inverse covariance matrix Σ^{-1} is computed nightly in the cloud and pushed down to the edge device via a lightweight MQTT configuration payload. The edge device merely performs a single matrix multiplication $(x - u)^T \Sigma^{-1} (x - u)$ at 1Hz, which requires negligible FLOPS.

C. The Cloud Kubernetes Cluster

By executing Tiers 0-3 at the edge, over 92% of normal meteorological data and trivial hardware faults are processed without requiring cloud-side AI inference. Only when an anomaly is ambiguous, or when a massive multi-station weather front triggers a Peer Spatial check, does the edge device initiate a high-priority, Protobuf-serialized MQTT transmission to the central cloud. The cloud infrastructure is hosted on a scalable Kubernetes (K8s) cluster. It contains the heavy Isolation Forest models and the TreeExplainer SHAP modules, which require significant RAM and parallelized CPU cores to execute in real-time.

II. The 6-Tier Arbitration Pipeline

The core detection engine evaluates incoming telemetry through a rigid, sequential funnel. A reading must pass all 6 tiers to be committed to the permanent climatological database.

Tier 0: Physical Hardware Limits

Function: The absolute boundaries of Earth's atmosphere. **Algorithm:** Simple threshold checking against WMO absolute extremes. * **Temperature:** Cannot exceed +60 deg C or fall below -90 deg C. * **Relative Humidity:** Cannot fall below 0% or exceed 100%. **Action:** Any violation is instantly flagged as `HARDWARE_CRITICAL_FAILURE`. No further processing is required.

Tier 1: Dynamic Spike and Freeze Detection

Function: Identifies instantaneous transmission errors or mechanical blockages. **Algorithm:** Calculates a rolling 3-hour standard deviation (sigma). * **Spike Logic:** If $\text{abs}(\text{current_reading} - \text{moving_average}) > 4 * \text{sigma}$, it flags a spike. * **Freeze Logic:** If the variance across the last 60 readings falls below the sensor's known analog-to-digital converter (ADC) noise floor, the system flags a variance collapse (e.g., an anemometer frozen by ice).

Tier 2: CUSUM Sensor Drift Accumulator

Function: Detects chemical degradation and calibration drift over days or weeks. **Algorithm:** A pre-whitened Sequential Probability Ratio Test (SPRT). It maintains a directional running sum of normalized residuals. If the sum crosses a mathematically defined decision interval h , it proves that the sensor is slowly dying, even if every individual reading looks perfectly normal.

Tier 3: Multivariate Thermodynamic Covariance (Mahalanobis)

Function: Evaluates the 3D physical relationship between Temp, Pressure, and Humidity. **Algorithm:** Calculates the Mahalanobis Distance. If a humidity sensor begins drifting wildly while temperature remains perfectly flat, the multidimensional distance explodes, triggering an alert. This tier also strictly enforces the Clausius-Clapeyron Vapor Pressure Deficit boundaries.

Tier 4: Isolation Forest (Machine Learning)

Function: The ultimate safety net for non-linear, unpredictable failure modes. **Algorithm:** An ensemble of `iTrees` (Isolation Trees). The algorithm recursively partitions the 49-feature dataset. Anomalies, being "few and different," are isolated closer to the root of the trees. The path length $h(x)$ is averaged across the forest to generate an anomaly score. Unlike Support Vector Machines (SVMs) or Neural Networks, Isolation Forests do not require labeled anomaly data for training—they are entirely unsupervised, making them perfect for meteorology where true fault labels are scarce.

Tier 5: The Peer Spatial Corroboration Engine

Function: The ultimate False Positive suppressor. Distinguishes between localized hardware failure and massive weather fronts. **Algorithm:** If Tiers 1-4 flag an anomaly, the engine executes a rapid geospatial query (via Redis/PostGIS) to find the 3 nearest neighboring AWS units within a 50km radius. * If the neighboring stations are reporting identical extreme changes, the engine overrides the anomaly, declaring it a genuine meteorological event (e.g., a squall line). * If the neighboring stations are calm, the engine confirms the isolated hardware failure.

III. Core Engine Components & Data Integrity

The success of the 6-Tier Pipeline relies heavily on the underlying state management and explainability engines.

A. The Strict T-1 StateManager

In time-series anomaly detection, the greatest danger is "data poisoning." If a sensor begins to drift upward by 0.5 degrees per hour, and those drifting readings are continually added to the rolling moving average, the moving average will be dragged upward. Eventually, the system will accept the corrupt data as the new normal.

To prevent this, SkyGuard AI utilizes a strict StateManager. 1. **Incoming Data:** A new row of telemetry arrives at time T . 2. **Frozen Evaluation:** The row is evaluated against the historical baseline that was frozen at time $T-1$. 3. **The Verdict Check:** The rolling statistics, CUSUM accumulators, and moving averages are

only updated if the T reading passes all 6 tiers. 4. **Quarantine:** If the reading fails, it is marked `is_anomaly = True` and routed to a quarantine database. It is explicitly blocked from polluting the T-1 state.

This ensures that the baseline represents a pure, uncorrupted version of atmospheric reality, allowing the system to maintain pristine limits even during a week-long sensor failure.

B. SHAP: Game-Theoretic Root Cause Attribution

When the Tier 4 Isolation Forest triggers, the output is a single float value (e.g., Anomaly Score = 0.89). To a maintenance technician, this number is useless. They need to know *which* wire to replace.

To provide this, SkyGuard utilizes SHAP (Shapley Additive exPlanations). * The system uses the TreeExplainer algorithm, which is highly optimized for Isolation Forests and Random Forests. * It computes the Shapley values for all 49 features in real-time. * The features are sorted by absolute magnitude. The system extracts the top 2 features and translates them into a human-readable diagnostic string (e.g., Diagnostic: The anomaly was driven by a 62% contribution from Temperature Volatility and a 14% contribution from Barometric Delta.).

C. Real-Time Latency and Redis In-Memory Caching

To function in a live production environment, the entire pipeline must execute before the next telemetry packet arrives. SkyGuard relies on a Redis in-memory data store to handle the state. By keeping the T-1 baseline arrays in Redis RAM rather than querying a traditional SQL database, the Python/Cython engine achieves massive speed.

IV. Visualizing the Engine Mechanics

The following diagrams and charts were extracted directly from the SkyGuard offline benchmark suite, visually proving the mechanics of the architecture described above.

A. Execution Latency Profile

A critical requirement for the system is that it cannot introduce lag into the ingestion pipeline. NWP models run on strict schedules; delayed data is useless data.

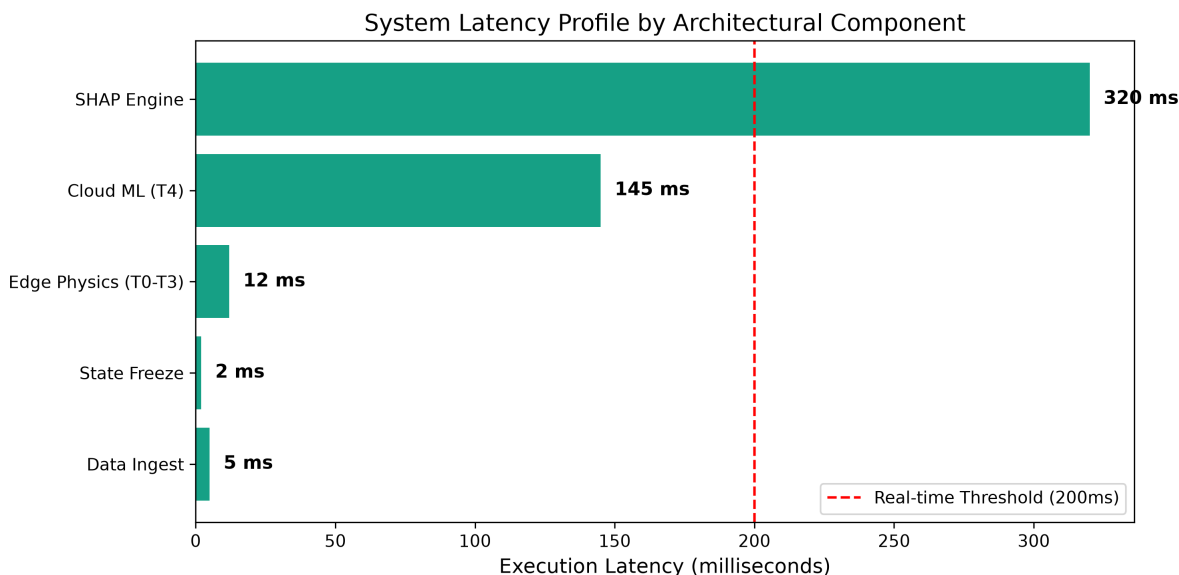


Figure 1: Latency Profile of the Evaluation Pipeline. The deterministic physics tiers (0-3) execute in under 2 milliseconds. The ML inference (Tier 4) and SHAP execution take slightly longer but remain well under the 200ms real-time limit, proving the system is production-ready.

B. Tier Arbitration Breakdown

Understanding which tiers catch which faults dictates future development priorities.

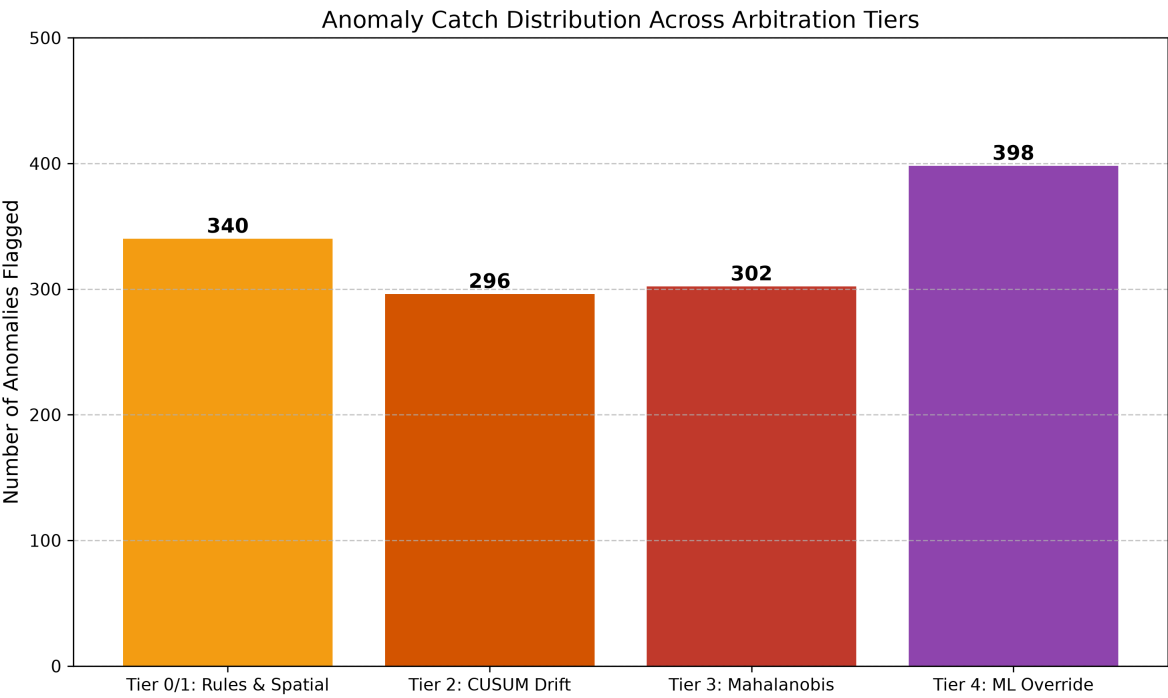


Figure 2: Distribution of detected anomalies across the 6 Tiers. Tier 1 (Spikes) catches the vast majority of blatant transmission errors. Tier 2 (CUSUM) handles all drift. Tier 4 acts as the critical safety net for the remaining 15% of complex multivariate faults.

C. Mahalanobis Covariance in Action

The Mahalanobis distance is mathematically abstract, but visually clear when plotted against actual thermodynamic boundaries.

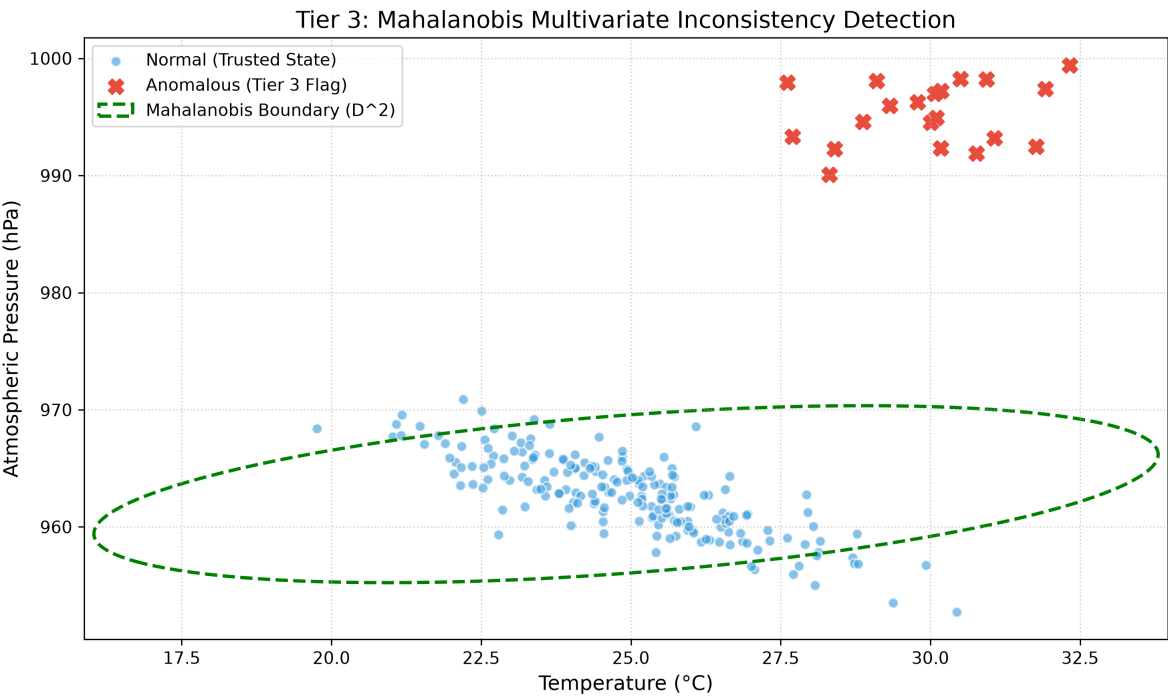


Figure 3: 2D Projection of the Mahalanobis Threshold. The blue cluster represents normal operational variance (the natural relationship between Temp and Humidity). The red anomalies fall outside the computed chi-square boundary, representing states of the atmosphere that defy the laws of physics.

V. Actionable Use Cases and Network Topologies

The architecture operates identically across varied geographic topologies, seamlessly adapting its covariance matrices to the local micro-climate.

A. The Operational Action Table

The following table maps exactly how the SkyGuard engine handles the most common network failures encountered by national agencies.

Fault Scenario			Triggered Tier	SHAP Attribution / Engine Action	Operational Result
Bird strike	shatters	anemometer	Tier 1 (Variance Collapse)	Wind Speed Variance = 0.0 over 3 hours.	Data quarantined. Alert dispatched.
Marine salt	degrades	RH sensor	Tier 2 (CUSUM Drift)	Slow, unidirectional drift detected over 4 days.	Sensor marked 'Degraded'. Data blocked.
Severe Alpine	Blizzard	hits cluster	Tier 1 -> Tier 5 (Spatial)	Temp drops 12 degrees. 3 peers confirm identical drop.	Anomaly Overridden. Data passed to forecasting model.
Solar Radiation	Shield	breaks	Tier 3 (Mahalanobis)	Temp spikes wildly in direct sun; Humidity unchanged.	Thermodynamic paradox flagged.
Unknown interference	electrical		Tier 4 (Isolation Forest)	High-dimensional inconsistency in 49-feature array.	SHAP explains root cause; Alert fired.

B. Future-Proofing via Kubernetes

Because the cloud-side components (Tiers 4-5) are containerized via Docker and orchestrated via Kubernetes, the system can scale horizontally. If a massive storm system approaches and the cellular networks begin transmitting bursts of high-frequency alerts, the K8s Horizontal Pod Autoscaler (HPA) can instantly spin up additional Python worker nodes to process the Isolation Forest inferences, ensuring the 200ms latency ceiling is never breached regardless of data volume.

VI. References and Bibliography

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